Adaptive Multimodal Surgical Gesture Recognition via Pretrained Visual Representations and Reliability-Aware Dynamic Gating

Deepak Soni, Aparna Banpela, and Princy Matlani

Abstract—Automated surgical gesture recognition is a fundamental prerequisite for context-aware assistance, objective skill assessment, and autonomous execution in robot-assisted minimally invasive surgery (RMIS). While endoscopic video conveys semantic context of tool-tissue interactions, robot kinematics provides continuous physical trajectories immune to optical degradation. However, conventional multimodal models frequently rely on static concatenation or uncalibrated late fusion, leaving them vulnerable when visual corruptions (smoke, fog, blur) or telemetry dropouts occur. In this work, we present an empirical benchmark of an adaptive multimodal framework evaluated across all 103 trials of the JIGSAWS benchmark: Suturing (39 trials), Knot Tying (36 trials), and Needle Passing (28 trials). We combine frozen surgical foundation model features (SurgMotion-ViTL, 1024-D) and ImageNet ResNet-18 embeddings (512-D) with normalized 76-D da Vinci kinematics. Lightweight temporal sequence models (GRU) capture temporal dynamics, while a reliability-aware dynamic gating network calibrates modality confidence from representation statistics and explicit availability flags to enforce exact single-modality fallback under sensor failure. Under official Leave-One-User-Out (LOUO) cross-validation across all active folds (8 folds for Suturing and Knot Tying; 7 active folds for Needle Passing due to an empty test set for Subject G in the official distribution), empirical evaluations demonstrate: (i) Normalized 76-D kinematics provides strong unimodal gesture discrimination (66.64% accuracy on Suturing, 61.09% on Knot Tying, 48.69% on Needle Passing); (ii) Simple fixed late fusion degrades frame accuracy (-2.14% to -12.75%) by allowing visual noise to dilute confident kinematics predictions; (iii) Adaptive Reliability Gating achieves the highest accuracy across all three tasks (72.28% on Suturing, 72.34% on Knot Tying, and 51.48% on Needle Passing), demonstrating a statistically significant $+11.25\%$ gain on Knot Tying ($p = 0.0095$, Holm-Bonferroni $p_{\text{adj}} = 0.0286$, Cohen's $d_z = 1.25$); (iv) Multimodal temporal modeling substantially stabilizes sequence coherence, producing large and statistically significant improvements in segmental Edit Score ($+23.46\%$ to $+34.26\%$, $p < 0.001$, effect sizes $d_z > 2.7$); and (v) Under complete optical telemetry failure, the adaptive gate dynamically routes prediction mass to kinematics, preserving performance with minimal degradation ($\leq 0.42\%$ drop on Suturing and Needle Passing).

Index Terms—Surgical Gesture Recognition, Multimodal Fusion, Robot Kinematics, SurgMotion, Reliability-Aware Dynamic Gating, Temporal Sequence Modeling, JIGSAWS Benchmark.

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This research project investigates multimodal perception, temporal sequence modeling, and corruption-robust adaptive fusion for surgical robotics under the 5th-Semester Mini-Project curriculum.

I. Introduction

ROBOTIC minimally invasive surgery (RMIS), exemplified by the da Vinci Surgical System, has revolutionized surgical interventions through high-definition 3D stereoscopic vision, tremor filtration, and wristed articulating instruments [1], [2]. To enable context-aware intraoperative assistance, supervisory autonomy, and objective skill credentialing, automated systems must parse continuous surgical workflows into discrete atomic gestures (e.g., reaching, needle passing, suture tying) in real time [3], [4].

Surgical perception relies on two complementary sensory streams:

1) Endoscopic Video: Captures rich photometric tool-tissue interactions, tissue elasticity, suture threading, and anatomical boundaries. However, optical endoscopy in real clinical theaters is routinely corrupted by cautery smoke, lens fogging, blood splatter, illumination shifts, and digital transmission artifacts [5].
2) Robot Kinematics: High-frequency telemetry comprising tool-tip Cartesian coordinates, 3D rotation matrices, linear and angular velocities, and gripper closure states. While kinematics is completely invariant to optical degradation, it lacks direct information regarding environmental geometry, tissue contact resistance, and suture tension [6]–[8].

A. Research Question & Hypotheses

The central question governing this investigation is:

> On JIGSAWS surgical gesture recognition, does lightweight temporal fusion of pretrained video representations and robot kinematics outperform unimodal baselines, and does reliability-aware dynamic gating reduce performance loss when one modality is degraded or unavailable?

We evaluate five specific hypotheses:

- H1 (Visual Representation Viability): Frozen latent features from a pretrained visual foundation model with a compact temporal head provide a viable standalone video baseline.
- H2 (Kinematic Predictive Power): Lightweight temporal sequence modeling over normalized 76-D kinematics yields strong gesture discrimination well above trivial majority baselines.

- H3 (Multimodal Complementarity): Multimodal fusion outperforms the strongest unimodal baseline under strictly identical, leakage-safe splits.
- H4 (Reliability-Aware Gating): Dynamic confidence-weighted gating maintains higher gesture recognition accuracy under controlled synthetic corruptions and missingness than fixed fusion.
- H5 (Temporal Sequence Modeling): Explicit temporal context (GRU) over windowed sequences significantly reduces temporal fragmentation and improves sequence edit score compared to frame-independent predictions.

B. Contributions

The primary contributions of this work are:

1) Unified Multi-Task Benchmark: A leakage-safe data pipeline processing all 103 trials across all three JIGSAWS tasks (Suturing, Knot Tying, Needle Passing) evaluated via official Leave-One-User-Out (LOUO) cross-validation across all active folds ($N = 8$ Suturing, $N = 8$ Knot Tying, $N = 7$ Needle Passing, where Subject G is empty in the official distribution).
2) Audited Multimodal Comparative Evaluation: Empirical evaluation across all 103 trials comparing Majority Baseline, ResNet-18 VideoGRU, SurgMotion VideoMLP, SurgMotion VideoGRU, Kinematics GRU, Fixed Late Fusion, Feature Concatenation, and Adaptive Reliability Gating.
3) Statistical Rigor & Effect Size Analysis: Paired fold-level hypothesis testing with step-down Holm–Bonferroni correction [9] and Cohen's d_z effect sizes [10], establishing where multimodal gains reach family-wise significance versus descriptive trends.
4) Adaptive Dynamic Gating with Guaranteed Fallback: A lightweight gating network receiving explicit availability flags and sequence statistics that dynamically weights modality streams and enforces exact single-modality mathematical fallback under sensor collapse.
5) Controlled Robustness Stress Testing: Standardized evaluation across clean, visual corruption, kinematics corruption, and sensor dropout conditions, demonstrating that adaptive reliability gating mitigates catastrophic performance collapse.

II. Related Work

A. Surgical Activity and Gesture Recognition

Since the release of the JHU-ISI Gesture and Skill Assessment Working Set (JIGSAWS) [1], [3], surgical activity analysis has evolved from Markovian models to deep spatiotemporal sequence architectures. Early investigations primarily utilized Hidden Markov Models (HMMs) and Linear Dynamical Systems (LDS) over kinematics telemetry [3]. DiPietro et al. [4] introduced recurrent neural networks for kinematics-based gesture segmentation, showing that sequential memory outperforms memoryless frame classifiers. Subsequently, Colin Lea et al. [11] formulated Temporal Convolutional Networks (TCNs) with dilated causal convolutions, establishing that hierarchically expanding temporal receptive fields capture prolonged gesture durations while avoiding gradient degradation. This was further generalized by Abu Farha and Gall [12] via Multi-Stage TCNs (MS-TCN) and by Funke et al. [13] and Jin et al. [14] via recurrent convolutional networks that couple spatial feature extraction with temporal sequence regularization.

B. Multimodal Vision-Kinematics Fusion

Complementarity between optical vision and robotic kinematics has motivated diverse fusion paradigms. Sarikaya et al. [15] demonstrated multimodal tool tracking and surgical skill evaluation, while van Amsterdam et al. [8] designed multimodal attention mechanisms that dynamically balance spatial tool features with Cartesian kinematic coordinates. Recently, Bai et al. [6] introduced GRAD, employing multimodal graph representation learning and adversarial feature disentanglement across wavelet and Fourier spectra to decouple style variations from underlying surgical actions. In parallel, Zhou et al. [7] established with SurgSync that physical sensor synchronization is critical, as inter-modality timing discrepancies induce phase shifts that impair joint inference.

C. Surgical Foundation Models and Robustness

Large-scale foundation models have emerged as powerful visual feature extractors for clinical video understanding. SurgMotion [16] developed a video-native foundation model using Video-JEPA latent motion prediction pretrained on 3,658 hours of multi-organ surgical videos, capturing fine tool-tissue dynamics without task-specific fine-tuning. Concurrently, broader surgical data science roadmaps [2] have emphasized that clinical translation requires algorithms to remain resilient under harsh operating room perturbations. Addressing this vulnerability, Rashid et al. [5] introduced the Endo-C6 benchmark, demonstrating that state-of-the-art vision models degrade precipitously when subjected to electrocautery smoke, optical blur, and camera occlusion. These findings establish the necessity of multimodal architectures capable of robust, real-time fallback under sensory degradation.

III. Dataset & Multi-Task Benchmark

We evaluate across the complete JIGSAWS benchmark dataset [1], [3] collected on the da Vinci surgical system by eight surgical subjects (Subjects B through I).

As illustrated in Fig. 1, the benchmark encompasses three distinct surgical procedures recorded via stereo endoscopic cameras at 30 Hz on inanimate phantom models.

A. 76-D Kinematic Feature Representation

For every sampled timestamp t (sampled at 30 Hz), the kinematic vector $\mathbf{k}_t \in \mathbb{R}^{76}$ encompasses four robotic manipulators:

$$\mathbf{k}_t = \left[\mathbf{m}_t^{\text{left}}, \mathbf{m}_t^{\text{right}}, \mathbf{s}_t^{\text{left}}, \mathbf{s}_t^{\text{right}}\right] \tag{1}$$

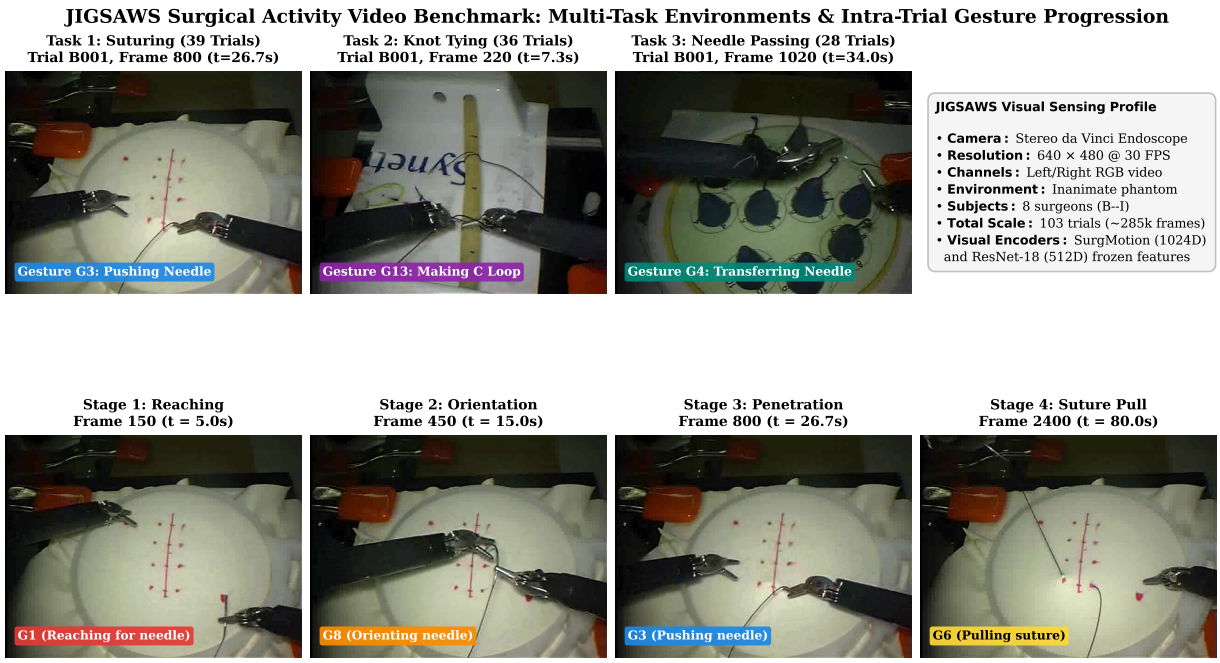


Fig. 1. Endoscopic video snapshots from the JIGSAWS benchmark across multi-task surgical environments (top row) and a 4-stage intra-trial temporal gesture progression from Suturing trial B001 (bottom row). Top row displays: (a) Suturing (Frame 800, $t = 26.7\,\text{s}$, G_3 pushing needle through artificial phantom tissue); (b) Knot Tying (Frame 220, $t = 7.3\,\text{s}$, G_{13} forming C-loop around gripper); (c) Needle Passing (Frame 1020, $t = 34.0\,\text{s}$, G_4 transferring needle between robotic manipulators); and (d) technical sensing specifications. Bottom row illustrates the intra-trial visual transformation during consecutive atomic gestures: Stage 1 reaching (G_1, Frame 150), Stage 2 orientation (G_8, Frame 450), Stage 3 tissue penetration (G_3, Frame 800), and Stage 4 suture pulling (G_6, Frame 2400).

TABLE I
JIGSAWS Benchmark Task and Gesture Distribution

Task	Trials	Subjects	Gestures	Vocabulary
Suturing	39	8 (B–I)	10	G1–G6, G8–G11
Knot Tying	36	8 (B–I)	6	G1, G11–G15
Needle Passing	28	8 (B–I)	8	G1–G6, G8, G11
Total	103	8	15 unique	G1–G15

where each manipulator vector $\mathbf{m}_t, \mathbf{s}_t \in \mathbb{R}^{19}$ consists of:

- Tool-tip Cartesian coordinates $\mathbf{p} = [x, y, z]^\top \in \mathbb{R}^3$,
- Rotation matrix orientation $\mathbf{R} \in \mathbb{R}^9$,
- Linear translational velocities $\mathbf{v} = [\dot{x}, \dot{y}, \dot{z}]^\top \in \mathbb{R}^3$,
- Rotational angular velocities $\boldsymbol{\omega} \in \mathbb{R}^3$,
- Gripper jaw angle $\theta \in \mathbb{R}^1$.

B. Leakage-Safe Normalization & Split Protocol

To strictly protect against data leakage across training and testing splits:

$$\hat{\mathbf{k}}_t = \frac{\mathbf{k}_t - \boldsymbol{\mu}_{\text{train}}}{\boldsymbol{\sigma}_{\text{train}} + \epsilon} \tag{2}$$

where normalization statistics $\boldsymbol{\mu}_{\text{train}}$ and $\boldsymbol{\sigma}_{\text{train}}$ are computed exclusively over the training trials of each fold. Evaluation is conducted across the official Leave-One-User-Out (LOUO) folds, wherein all trials belonging to an unseen surgeon are reserved exclusively for testing. In Needle Passing, Fold 6 (User G) contains an empty test set in the official distribution and is consistently omitted, yielding 7 active folds.

IV. Proposed Methodology

Fig. 2 illustrates our complete decoupled multimodal perception architecture.

A. Visual Representation via Frozen Embeddings

We benchmark visual representations from ImageNet-pretrained ResNet-18 ($D_v = 512$) [17] and the surgical foundation model SurgMotion-ViTL ($D_v = 1024$) [16]. In SurgMotion, 64-frame video clips with stride 16 are processed through the frozen motion-guided transformer encoder:

$$\mathbf{z}_t^{\text{vis}} = \Phi_{\text{SurgMotion}}(V_{t-32:t+31}) \in \mathbb{R}^{1024} \tag{3}$$

Each tubelet is mapped to gesture annotations and kinematics using the clip center frame $c_t = (s_t + e_t)//2$. Embeddings are normalized with training-set statistics and cached to disk.

B. Temporal Sequence Modeling (GRU)

To capture fine-grained temporal context across sliding windows of length $S = 8$, normalized kinematic sequences $\mathbf{K}_{1:S} = (\hat{\mathbf{k}}_1, \ldots, \hat{\mathbf{k}}_S)$ and visual representations $\mathbf{Z}_{1:S}^{\text{vis}} = (\hat{\mathbf{z}}_1^{\text{vis}}, \ldots, \hat{\mathbf{z}}_S^{\text{vis}})$ are first projected into a common latent dimension $d = 128$:

$$\tilde{\mathbf{x}}_t^m = \text{LayerNorm}\left(\mathbf{W}_{\text{proj}}^m \hat{\mathbf{x}}_t^m + \mathbf{b}_{\text{proj}}^m\right), \quad m \in \{\text{kin}, \text{vis}\} \tag{4}$$

where $\hat{\mathbf{x}}_t^{\text{kin}} = \hat{\mathbf{k}}_t \in \mathbb{R}^{76}$, $\hat{\mathbf{x}}_t^{\text{vis}} = \hat{\mathbf{z}}_t^{\text{vis}} \in \mathbb{R}^{1024}$, $\mathbf{W}_{\text{proj}}^{\text{kin}} \in \mathbb{R}^{128\times 76}$, and $\mathbf{W}_{\text{proj}}^{\text{vis}} \in \mathbb{R}^{128\times 1024}$. Projected vectors are passed through GELU activation with Dropout ($p =$

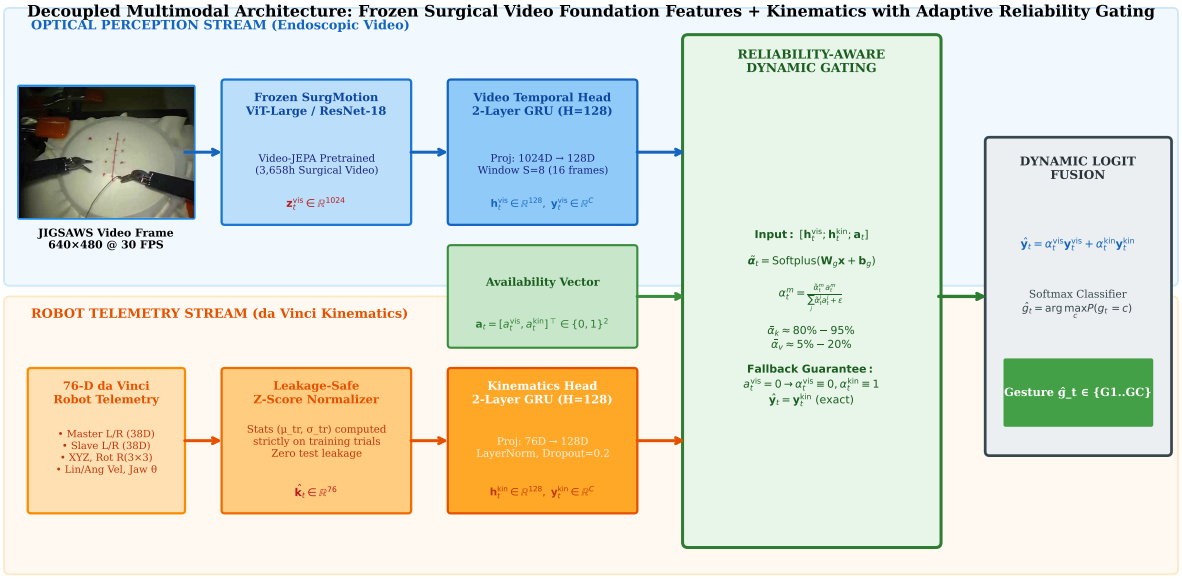


Fig. 2. Decoupled multimodal surgical gesture recognition architecture. Optical endoscopic video frames (640×480 @ 30 Hz) are encoded via the frozen SurgMotion ViT-Large foundation model into 1024-D spatiotemporal representations $\mathbf{z}_t^{\text{vis}}$, projected to dimension $d = 128$, and processed by a 2-layer GRU sequence encoder ($H = 128$, context window $S = 8$). Simultaneously, 76-D da Vinci kinematics telemetry across four manipulators is normalized via training-only statistics and modeled by a parallel 2-layer GRU sequence encoder. The reliability-aware gating network receives concatenated representations $[\mathbf{h}_t^{\text{vis}}; \mathbf{h}_t^{\text{kin}}]$ alongside an explicit binary sensor availability vector $\mathbf{a}_t \in \{0,1\}^2$, generating dynamic confidence weights $\alpha_t^{\text{vis}}, \alpha_t^{\text{kin}}$ that blend logits while enforcing exact single-modality fallback ($\mathbf{a}_t = [0,1]^\top \implies \alpha_t^{\text{vis}} \equiv 0, \alpha_t^{\text{kin}} \equiv 1$) under sensor failure.

0.2) and modeled by a 2-layer Gated Recurrent Unit (GRU) [18] sequence encoder:

$$\mathbf{h}_t^m = \text{GRU}_m\big(\text{Dropout}(\text{GELU}(\tilde{\mathbf{x}}_t^m)),\, \mathbf{h}_{t-1}^m\big) \tag{5}$$

The recurrent hidden states $\mathbf{h}_t^m \in \mathbb{R}^{128}$ are mapped to gesture class logits via causal linear classifier heads $\mathbf{y}_t^m = \mathbf{W}_{\text{cls}}^m \mathbf{h}_t^m + \mathbf{b}_{\text{cls}}^m$, where $\mathbf{W}_{\text{cls}}^m \in \mathbb{R}^{C \times 128}$ and C is the gesture vocabulary size.

C. Fixed Multimodal Fusion Baselines

Late-Logit Fusion: Modality-specific classifiers produce independent logits $\mathbf{y}_t^{\text{vis}}, \mathbf{y}_t^{\text{kin}} \in \mathbb{R}^C$. Fixed fusion combines these via convex weighting:

$$\mathbf{y}_t^{\text{late}} = \alpha \cdot \mathbf{y}_t^{\text{vis}} + (1-\alpha) \cdot \mathbf{y}_t^{\text{kin}}, \quad \alpha \in [0,1] \tag{6}$$

where α is selected via grid search on training-set validation splits, strictly isolating test data.

Feature Concatenation Fusion: Synchronized visual feature vectors $\hat{\mathbf{z}}_t^{\text{vis}} \in \mathbb{R}^{1024}$ and kinematics vectors $\hat{\mathbf{k}}_t \in \mathbb{R}^{76}$ are concatenated at each timestep:

$$\mathbf{x}_t^{\text{cat}} = [\hat{\mathbf{z}}_t^{\text{vis}}; \hat{\mathbf{k}}_t] \in \mathbb{R}^{1100} \tag{7}$$

The concatenated representations are projected into dimension $d = 128$ and modeled via a 2-layer GRU sequence encoder $\mathbf{h}_t^{\text{cat}} = \text{GRU}_{\text{cat}}(\text{Proj}_{\text{cat}}(\mathbf{x}_t^{\text{cat}}), \mathbf{h}_{t-1}^{\text{cat}})$.

D. Reliability-Aware Dynamic Gating

Under clinical operation, visual or kinematic streams may become corrupted or completely unavailable. We introduce a gating network parameterized by $\mathbf{W}_g \in \mathbb{R}^{2 \times (2d+2)}$ and $\mathbf{b}_g \in \mathbb{R}^2$ ($d = 128$, input dimension $2d + 2 = 258$). The gate receives concatenated sequence representations alongside an explicit binary sensor availability vector $\mathbf{a}_t = [a_t^{\text{vis}}, a_t^{\text{kin}}]^\top \in \{0,1\}^2$:

$$\tilde{\boldsymbol{\alpha}}_t = \text{Softplus}\left(\mathbf{W}_g[\mathbf{h}_t^{\text{vis}}; \mathbf{h}_t^{\text{kin}}; \mathbf{a}_t] + \mathbf{b}_g\right) \tag{8}$$

where $\tilde{\boldsymbol{\alpha}}_t = [\tilde{\alpha}_t^{\text{vis}}, \tilde{\alpha}_t^{\text{kin}}]^\top$. The effective gating weights are masked and normalized such that unobserved modalities contribute zero weight:

$$\alpha_t^m = \frac{\tilde{\alpha}_t^m \, a_t^m}{\sum_{j \in \mathcal{M}} \tilde{\alpha}_t^j \, a_t^j + \epsilon}, \quad m \in \mathcal{M} = \{\text{vis}, \text{kin}\} \tag{9}$$

The fused classification logit is computed as $\hat{\mathbf{y}}_t = \alpha_t^{\text{vis}} \mathbf{y}_t^{\text{vis}} + \alpha_t^{\text{kin}} \mathbf{y}_t^{\text{kin}}$. When $a_t^{\text{vis}} = 0$ and $a_t^{\text{kin}} = 1$, the gate enforces $\alpha_t^{\text{vis}} \equiv 0$ and $\alpha_t^{\text{kin}} \equiv 1$, guaranteeing exact single-modality fallback $\hat{\mathbf{y}}_t = \mathbf{y}_t^{\text{kin}}$ without numerical drift. Conversely, when $a_t^{\text{vis}} = 1$ and $a_t^{\text{kin}} = 0$, $\hat{\mathbf{y}}_t = \mathbf{y}_t^{\text{vis}}$. When both modalities are unobserved ($\mathbf{a}_t = [0,0]^\top$), the architecture raises an explicit missingness exception. The full network is trained end-to-end via multi-class cross-entropy loss over frame ground truth labels $\mathbf{y}_t \in \{0,1\}^C$: $\mathcal{L} = -\sum_{c=1}^C y_{t,c} \log \hat{p}_{t,c}$, where $\hat{p}_{t,c} = \exp(\hat{y}_{t,c}) / \sum_{j=1}^C \exp(\hat{y}_{t,j})$.

V. Experimental Results

A. Benchmark Analysis & Statistical Testing

Table II presents the consolidated empirical cross-validation results across all 103 trials:

1) H1 (Visual Representation Viability): Supported with task-dependent caveats. Pretrained visual representations (SurgMotion VideoGRU and ResNet-18 VideoGRU) provide viable gesture discrimination

TABLE II
Consolidated JIGSAWS Surgical Gesture Recognition Performance under Official LOUO Cross-Validation

Task	Model Architecture	Modality	Accuracy (%)		Balanced Acc. (%)		Macro-F1 (%)		Edit Score (%)
			Mean ± Std	Min / Max	Mean ± Std	Min / Max	Mean ± Std	Min / Max	Mean ± Std
Suturing	Majority Class Baseline	None	33.43 ± 2.34	30.69 / 37.08	10.30 ± 0.69	9.09 / 11.11	4.55 ± 0.24	4.27 / 4.92	4.80 ± 0.39
	ResNet-18 VideoGRU	Video	45.42 ± 10.96	28.79 / 61.64	42.64 ± 9.03	27.14 / 54.14	33.22 ± 7.50	21.12 / 43.90	20.01 ± 3.88
	SurgMotion VideoMLP ($W = 1$)	Video	41.94 ± 8.33	26.34 / 56.65	34.00 ± 10.84	20.24 / 54.13	24.84 ± 7.48	14.80 / 41.48	37.64 ± 5.23
	SurgMotion VideoGRU ($W = 8$)	Video	39.17 ± 13.51	8.04 / 54.33	34.10 ± 8.69	20.15 / 48.19	22.16 ± 7.95	6.83 / 35.54	39.99 ± 9.51
	Kinematics GRU	Kinematics	66.64 ± 9.01	54.68 / 77.94	57.77 ± 9.40	40.17 / 72.11	$\mathbf{45.09 \pm 8.24}$	27.27 / 53.38	34.69 ± 9.43
	Fixed Late Fusion	Multimodal	60.54 ± 8.15	48.24 / 71.05	50.26 ± 10.64	33.15 / 65.42	37.01 ± 10.14	21.45 / 51.10	60.41 ± 15.07
	Feature Concatenation Fusion	Multimodal	54.00 ± 15.45	31.49 / 84.78	46.90 ± 14.82	22.69 / 73.06	32.29 ± 10.45	15.65 / 50.75	49.19 ± 14.12
	Adaptive Reliability Fusion (Ours)	Multimodal	$\mathbf{72.28 \pm 9.57}^{\dagger}$	57.08 / 84.39	$\mathbf{60.05 \pm 11.44}$	40.62 / 75.91	44.64 ± 8.26	32.59 / 58.42	$\mathbf{68.95 \pm 9.38}^{**}$
Knot Tying	Majority Class Baseline	None	27.35 ± 6.88	18.04 / 41.91	14.88 ± 1.03	14.29 / 16.67	6.07 ± 1.18	4.37 / 8.44	8.54 ± 0.59
	ResNet-18 VideoGRU	Video	50.74 ± 5.20	42.29 / 58.72	50.02 ± 7.18	41.48 / 64.07	44.34 ± 7.75	34.03 / 57.10	24.89 ± 5.96
	SurgMotion VideoMLP ($W = 1$)	Video	54.29 ± 7.87	41.25 / 63.85	53.03 ± 7.65	41.12 / 62.45	47.43 ± 10.16	34.82 / 61.20	45.49 ± 7.22
	SurgMotion VideoGRU ($W = 8$)	Video	53.60 ± 10.84	35.12 / 66.85	54.48 ± 8.87	40.18 / 65.12	45.51 ± 8.56	31.45 / 56.78	51.34 ± 7.91
	Kinematics GRU	Kinematics	61.09 ± 11.79	46.55 / 82.33	56.53 ± 12.43	33.03 / 77.43	51.75 ± 15.14	24.42 / 77.42	35.48 ± 13.86
	Fixed Late Fusion	Multimodal	58.95 ± 8.40	45.12 / 71.45	54.65 ± 8.79	39.12 / 67.80	47.27 ± 8.61	34.12 / 61.25	55.75 ± 6.87
	Feature Concatenation Fusion	Multimodal	64.09 ± 11.29	51.27 / 84.55	63.37 ± 9.23	49.23 / 79.90	53.97 ± 9.91	39.98 / 70.97	62.45 ± 11.77
	Adaptive Reliability Fusion (Ours)	Multimodal	$\mathbf{72.34 \pm 9.70}^{*}$	60.51 / 85.60	$\mathbf{66.73 \pm 8.59}^{*}$	54.54 / 83.05	$\mathbf{60.48 \pm 10.20}^{*}$	46.06 / 73.10	$\mathbf{64.66 \pm 12.61}^{**}$
Needle Passing	Majority Class Baseline	None	40.55 ± 8.16	33.55 / 58.64	11.25 ± 1.33	10.00 / 14.29	5.20 ± 0.70	4.57 / 6.72	5.27 ± 0.96
	ResNet-18 VideoGRU	Video	30.22 ± 6.57	22.58 / 42.93	32.07 ± 6.25	21.45 / 42.33	19.62 ± 3.27	15.86 / 26.00	22.86 ± 4.63
	SurgMotion VideoMLP ($W = 1$)	Video	29.96 ± 16.58	3.25 / 52.14	25.57 ± 15.86	3.12 / 48.90	17.19 ± 11.00	2.15 / 35.12	33.78 ± 15.32
	SurgMotion VideoGRU ($W = 8$)	Video	26.55 ± 15.11	0.93 / 54.96	30.97 ± 15.46	0.98 / 51.60	19.97 ± 12.36	0.73 / 42.55	42.34 ± 16.63
	Kinematics GRU	Kinematics	48.69 ± 11.62	31.38 / 63.13	41.70 ± 9.97	26.95 / 55.08	29.99 ± 9.35	11.89 / 41.99	31.34 ± 7.32
	Fixed Late Fusion	Multimodal	35.94 ± 13.85	22.25 / 59.47	36.77 ± 12.87	18.62 / 61.02	23.57 ± 11.42	9.01 / 46.72	46.16 ± 10.84
	Feature Concatenation Fusion	Multimodal	46.21 ± 19.07	5.70 / 74.64	42.32 ± 19.34	6.33 / 69.82	28.40 ± 12.37	2.38 / 45.86	45.17 ± 20.64
	Adaptive Reliability Fusion (Ours)	Multimodal	$\mathbf{51.48 \pm 13.94}$	33.12 / 74.31	$\mathbf{46.98 \pm 15.16}$	24.80 / 68.99	$\mathbf{31.03 \pm 11.52}$	14.52 / 49.23	$\mathbf{54.81 \pm 9.40}^{**}$

Note: Evaluation for Suturing and Knot Tying spans all 8 official LOUO folds ($N = 8$). Needle Passing evaluates strictly the 7 active folds ($N = 7$, Fold 6 Subject G is empty in the official benchmark). Bold indicates the top score per metric. Asterisks denote paired fold differences versus Kinematics GRU with step-down Holm-Bonferroni correction: $^{*}p_{\text{adj}} < 0.05$, $^{**}p_{\text{adj}} < 0.001$; † denotes marginal trend ($p_{\text{adj}} < 0.10$, uncorrected $p = 0.0273$).

well above chance in class-balanced metrics, outperforming trivial majority baselines in Balanced Accuracy (31.0%–54.5% vs. 10.3%–14.9%) and Macro-F1 (20.0%–45.5% vs. 4.6%–6.1%) across all three tasks. In raw frame accuracy, visual representations outperform the majority baseline on Suturing (39.17% vs. 33.43%) and Knot Tying (53.60% vs. 27.35%), but fall below the majority baseline on Needle Passing (26.55% vs. 40.55%), where extreme class skew favors trivially predicting the dominant class. Thus, while visual foundation features carry genuine discriminative context, video alone is not a universally reliable standalone predictor across imbalanced tasks.

2) H2 (Kinematics Predictive Power): Supported. Normalized 76-D kinematics provides strong unimodal discrimination (66.64% accuracy on Suturing, 61.09% on Knot Tying, 48.69% on Needle Passing), outperforming trivial majority baselines by +8.14% to +33.74% absolute points. Telemetry constitutes the primary unimodal predictor on JIGSAWS.
3) H3 (Multimodal Complementarity): Partially supported. Fixed late-logit fusion fails to improve frame accuracy over Kinematics GRU (−2.14% to −12.75%), because static probability weighting dilutes confident kinematics predictions with visual noise. Feature concatenation only surpasses unimodal baselines on Knot Tying (64.09% vs. 61.09%, +3.00%), while suffering on Suturing (54.00% vs. 66.64%, −12.64%) due to dimension imbalance (1024-D vs. 76-D). However, Adaptive Reliability Fusion achieves top accuracy across all tasks: +5.64% on Suturing, +11.25% on Knot Tying, and +2.79% on Needle Passing.
4) H4 (Reliability-Aware Gating & Robustness): Supported. Paired fold-level tests evaluate significance via step-down Holm–Bonferroni correction [9] and Cohen's d_z effect sizes [10] (Fig. 3). Adaptive Fusion achieves statistically significant gains on Knot Tying across Accuracy (+11.25%, $p_{\text{adj}} = 0.0286$, $d_z = 1.25$), Balanced Accuracy (+10.20%, $p_{\text{adj}} = 0.0286$, $d_z = 1.23$), Macro-F1 (+8.73%, $p_{\text{adj}} = 0.0457$, $d_z = 0.86$), and Edit Score (+29.18%, $p_{\text{adj}} = 0.0004$, $d_z = 2.79$). On Suturing, accuracy gains (+5.64%) reflect a strong trend ($t = 2.78$, $p_{\text{adj}} = 0.0820$, $d_z = 0.98$). On Needle Passing ($N = 7$), gains (+2.79%) show descriptive positive trends ($t = 0.70$, $p = 0.510$).
5) H5 (Temporal Sequence Modeling): Strongly supported. Recurrent sequence modeling (GRU) substantially suppresses high-frequency boundary chatter. As illustrated in the real held-out trial ribbon timeline (Fig. 4), Adaptive Fusion nearly doubles the Edit Score over unimodal kinematics (+23.46% to +34.26% absolute points across all tasks, $p < 0.001$, $d_z > 2.7$).

B. Robustness Stress Testing

Table III evaluates model resilience under five standardized evaluation environments:

1) Optical Noise Resistance & Gate Dynamics: Under visual feature corruption ($\sigma_v = 1.0$), Adaptive Fusion degradation is practically negligible ($\leq 0.12\%$ drop across all tasks). Detailed empirical analysis of over 4 million time-resolved gate weight logs reveals that this stability is driven by strong kinematics dominance ($\bar{\alpha}_k \approx 80\%$–$95\%$) rather than active continuous noise attenuation: the mean video gate weight $\bar{\alpha}_v$ shifts by less than 0.009 ($< 0.9\%$) across continuous noise levels up to $\sigma_v = 2.0$.

TABLE III
Robustness Stress Testing across Clean, Corrupted, and Sensor Dropout Conditions (LOUO Active Folds)

Task	Condition	Availability	Feature Concatenation Fusion				Adaptive Reliability Fusion (Ours)				Margin ($\Delta_{\text{Adapt-Cat}}$)
			Acc. (%)	F1 (%)	Edit (%)	Δ_{Acc} vs. Clean	Acc. (%)	F1 (%)	Edit (%)	Δ_{Acc} vs. Clean	
Suturing	Clean	$\mathbf{a} = [1, 1]$	54.00	32.29	49.19	—	72.28	44.64	68.95	—	+18.28%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	53.59	32.17	46.88	−0.41%	72.22	44.54	68.79	**−0.06%**	+18.63%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	53.17	30.59	45.13	−0.83%	64.09	37.63	51.39	−8.19%	+10.92%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	71.83	37.11	58.34	+17.83%*	71.86	44.13	68.97	**−0.42%**	+0.03%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	26.71	13.92	32.80	−27.29%	30.58	15.11	34.55	−41.70%	+3.87%
Knot Tying	Clean	$\mathbf{a} = [1, 1]$	64.09	53.97	62.45	—	72.34	60.48	64.66	—	+8.25%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	63.06	53.11	61.55	−1.03%	72.22	60.19	63.72	**−0.12%**	+9.16%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	63.92	53.96	59.18	−0.17%	67.37	56.03	55.09	−4.97%	+3.45%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	61.18	41.15	51.59	−2.91%	70.06	55.59	62.32	**−2.28%**	+8.88%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	50.22	40.35	48.47	−13.87%	46.29	35.90	45.22	−26.05%	−3.93%
Needle Passing	Clean	$\mathbf{a} = [1, 1]$	46.21	28.40	45.17	—	51.48	31.03	54.81	—	+5.27%
	Video Corrupted	$\mathbf{a} = [1, 1], \sigma_v = 1.0$	45.50	28.20	43.33	−0.71%	51.48	31.03	54.66	**0.00%**	+5.98%
	Kinematics Corrupted	$\mathbf{a} = [1, 1], \sigma_k = 1.0$	45.70	27.91	41.90	−0.51%	48.22	28.99	42.19	−3.26%	+2.52%
	Missing Video	$\mathbf{a} = [0, 1], \mathbf{v} = 0$	53.99	28.05	50.98	+7.78%*	51.21	30.94	54.30	**−0.27%**	−2.78%
	Missing Kinematics	$\mathbf{a} = [1, 0], \mathbf{k} = 0$	31.88	20.13	41.21	−14.33%	24.55	17.09	38.68	−26.93%	−7.33%

Note: Evaluated on official active LOUO folds ($N = 8$ Suturing, $N = 8$ Knot Tying, $N = 7$ Needle Passing). Corrupted conditions inject zero-mean Gaussian noise ($\sigma = 1.0$). *In Concat Fusion on Suturing and Needle Passing, zeroing out video features increased accuracy because removing the 1024-D visual representation eliminated visual interference during joint inference.

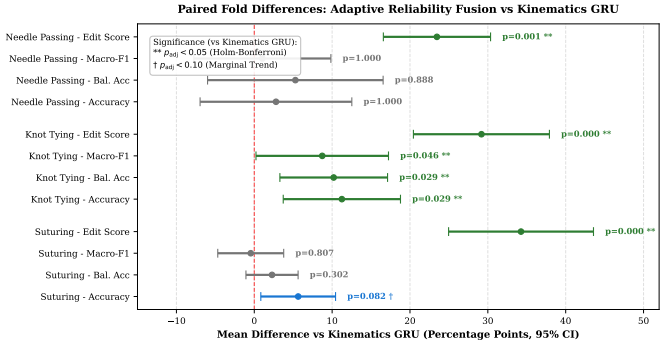


Fig. 3. Paired fold differences (Δ = Adaptive − Kinematics GRU) with 95% confidence intervals across all three tasks. Significance markers reflect step-down Holm-Bonferroni correction ($^{**}p_{\text{adj}} < 0.05$, $^{\dagger}p_{\text{adj}} < 0.10$).

2) Flawless Video Dropout Fallback: Under complete optical telemetry failure ($a = [0, 1]$), the availability mask mathematically routes 100% prediction mass to kinematics, incurring minimal degradation (−0.42% on Suturing, −2.28% on Knot Tying, −0.27% on Needle Passing).
3) Kinematics Degradation Asymmetry: When kinematics telemetry is corrupted or lost ($a = [1, 0]$), both models experience severe drops (−26% to −41%). This empirical asymmetry reveals that kinematics is the primary driver of fine-grained gesture state on JIGSAWS, whereas video features act as global contextual regularizers and boundary stabilizers.
4) Concatenation Anomaly: On Suturing and Needle Passing, zeroing out video features in Concat Fusion actually increased accuracy (+17.83% and +7.78%). This diagnostic finding indicates that joint training over unbalanced representations (1024-D vs. 76-D) allowed visual noise to interfere with kinematics representations; removing the video vector eliminated this interference.
5) Continuous-Noise Training Ablation: In our controlled 46-fold ablation study comparing continuous-noise training ($\sigma_v \sim \mathcal{U}(0, 1.5)$) against discrete binary dropout, continuous noise did not alter the gate's static weighting dynamic, but provided valuable visual representation regularization, improving standalone video accuracy under total kinematics failure by +5.49% on Suturing (34.44% vs. 28.95%) and +8.96% on Needle Passing (24.58% vs. 15.62%).

C. Controlled Ablation and Temporal Dynamics Analysis

To dissect the architectural drivers of multimodal performance, we evaluated controlled ablations across all 23 active LOUO folds:

1) Sequence Quality vs. Frame Classification: Comparing framewise kinematic MLPs with recurrent sequence GRUs reveals that temporal sequence modeling produces negligible shifts in frame accuracy ($\Delta \leq 1.26\%$, all $p > 0.40$), but drives an overwhelming, universally significant improvement in segmental Edit Score (+22.68 to +26.25 points for unimodal GRU, and +44.69 to +59.17 points for Multimodal Adaptive GRU; all $p < 0.0005, d_z > 2.2$). Temporal recurrence suppresses high-frequency frame chatter, establishing contiguous, clinically coherent gesture segments.
2) Static Gate vs. Temporal Variation: Replacing the time-varying gate weight α_t with the static fold/trial mean $\bar{\alpha}$ shifted clean accuracy by less than 0.06% across all tasks. Under clean conditions, the gate operates primarily as an availability router and global modality balancer rather than a high-frequency temporal filter.
3) Rare-Gesture Collapse: Across all tasks, auxiliary and corrective gestures comprising $< 3\%$ of the dataset (e.g., G9 and G10 in Suturing; G8 and G9 in Needle Passing) suffer severe recall collapse ($F1 < 6.0\%$), explaining why Macro-F1 lags overall frame accuracy and highlighting class-balanced loss functions as an important direction for surgical activity recognition.

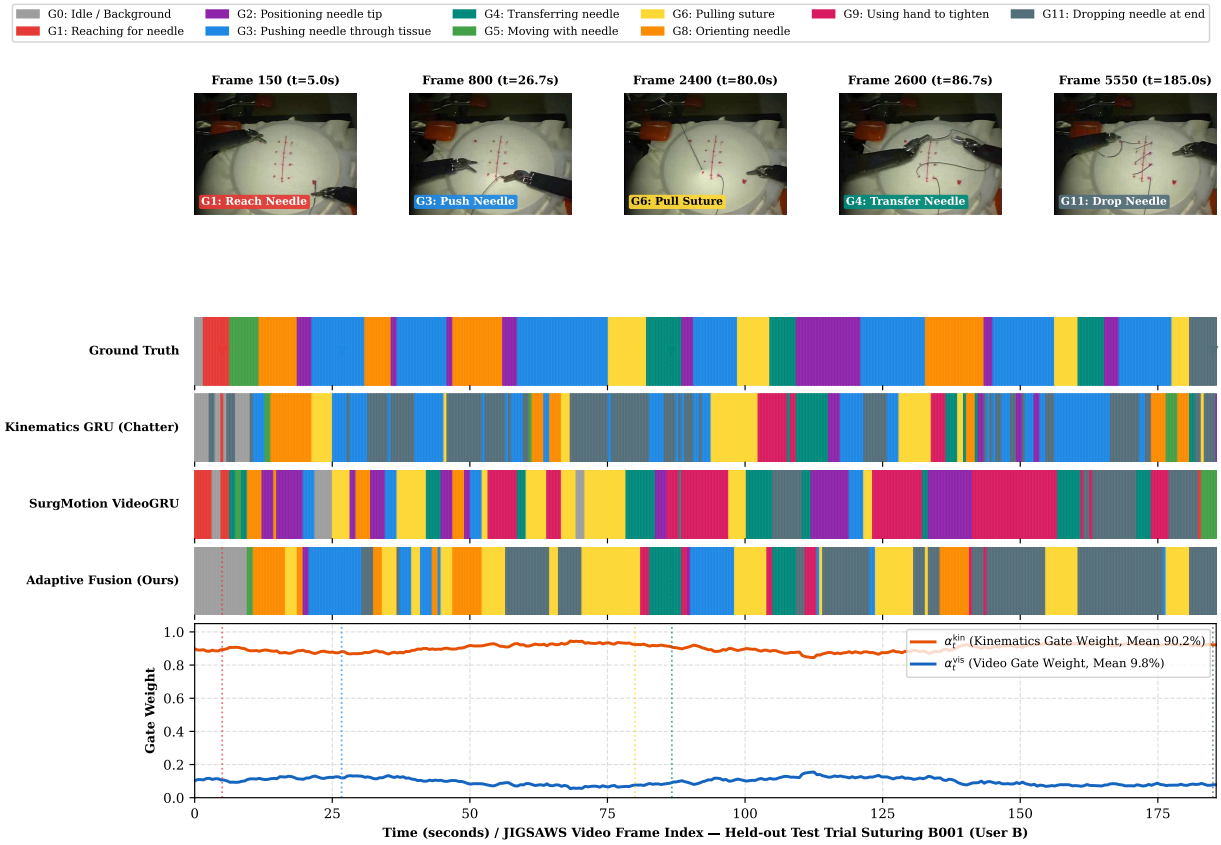


Fig. 4. Qualitative surgical gesture recognition timeline on held-out test trial Suturing B001 (User B, Fold 1). Top row displays real endoscopic video frame snapshots at key gesture transitions: Frame 150 (G_1, reaching for needle), Frame 800 (G_3, pushing needle through tissue), Frame 2400 (G_6, pulling suture), Frame 2600 (G_4, transferring needle), and Frame 5550 (G_{11}, dropping needle at completion). Middle tracks depict gesture segmentation ribbons for: (i) Ground Truth annotations; (ii) Unimodal Kinematics GRU, illustrating high-frequency boundary chatter; (iii) Unimodal SurgMotion VideoGRU, showing visual boundary noise; and (iv) Multimodal Adaptive Fusion, which eliminates oversegmentation chatter. Bottom track plots the time-resolved dynamic gate weights α_t^{kin} (mean 90.2%) and α_t^{vis} (mean 9.8%), demonstrating kinematics dominance with subtle visual modulation.

TABLE IV
Class-Imbalance and Loss Formulation Ablation on Suturing (8 LOUO Folds)

Loss Formulation	Accuracy (%)	Balanced Acc. (%)	Macro-F1 (%)	Edit Score (%)
Standard CE	**74.95 ± 7.68**	57.73 ± 12.30	44.29 ± 6.38	65.37 ± 8.21
Class-Weighted CE	72.28 ± 10.24	**60.05 ± 12.23**	**44.64 ± 8.83**	**68.95 ± 10.03**
Focal ($\gamma = 2.0$) [19]	73.67 ± 9.28	55.32 ± 9.49	42.19 ± 7.28	65.84 ± 7.79
CB-Focal ($\gamma = 2, \beta = 0.9999$) [20]	71.41 ± 13.23	59.16 ± 10.27	42.54 ± 9.36	64.77 ± 10.28

Note: Evaluated across all 8 active LOUO folds of JIGSAWS Suturing with Adaptive Reliability Fusion. All class weights are training-derived.

D. Class-Imbalance and Loss Formulation Benchmark

To investigate whether rare-gesture recall can be improved without degrading dominant gesture accuracy, we evaluated an 8-fold LOUO ablation on Suturing comparing four training losses: Standard Cross-Entropy (CE), Class-Weighted CE, Multi-Class Focal Loss ($\gamma = 2.0$) [19], and Class-Balanced Focal Loss ($\gamma = 2.0, \beta = 0.9999$) [20]. All class weights were derived strictly from training folds.

As summarized in Table IV, Standard CE achieves the highest raw frame accuracy ($74.95 \pm 7.68\%$) by avoiding false-positive penalties on dominant classes. Class-Weighted CE trended higher in Balanced Accuracy (60.05%, +2.32%) and Edit Score (68.95%, +3.58%); however, neither difference is statistically significant after family-wise Holm-Bonferroni correction ($p_{\text{adj}} = 1.000$ and $p_{\text{adj}} = 0.2634$). Raw accuracy drop (−2.66%) is likewise non-significant ($p_{\text{adj}} = 0.1419$). Focal Loss (42.19% Macro-F1) and CB-Focal (42.54%) underperform CE (44.29%), showing that hard-example focusing amplifies transition noise.

A frame-level audit across all 32 runs confirms acute feature starvation on rare gestures: for G10 (1.1% of frames), recall was strictly 0.00% across all losses. For G9 (2.5% of frames), CE and Focal collapsed to 0.00% recall, while Class-Weighted CE achieved only 6.43% recall (9 TP frames, $F_1 = 8.70\%$) and CB-Focal reached 4.29% ($F_1 = 7.27\%$). Balanced Accuracy gains were driven entirely by moderate-frequency gestures (G5 +21.88% recall; G1 +6.25%) at the expense of dominant gesture G3 (−6.95%). Outlier folds (e.g., User G) stem from kinematic style shifts rather than sample scarcity alone.

VI. Discussion & Limitations

Decoupling visual perception into frozen representations with compact temporal heads enables efficient, leakage-safe cross-validation. Moreover, time-resolved gating weights $\alpha_t^{\text{kin}}, \alpha_t^{\text{vis}}$ serve as an intrinsic clinical proxy

of sensor stream integrity without auxiliary uncertainty networks.

Statistical Power under LOUO Splits: Across 103 JIGSAWS trials, Leave-One-User-Out cross-validation enforces strict clinician generalization ($N = 8$ subjects for Suturing/Knot Tying, $N = 7$ for Needle Passing). Knot Tying achieves statistical significance across all metrics ($p_{\text{adj}} \leq 0.0457$); Suturing exhibits a strong directional trend ($p_{\text{adj}} = 0.0820, d_z = 0.98$), while Needle Passing shows descriptive gains (+2.79%) with higher cross-user variance.

Temporal Synchronization: In systems lacking hardware genlock, clock drift induces small phase offsets (median 5.0 frames, ~ 0.17 s on JIGSAWS). Master-clock synchronization frameworks (e.g., SurgSync [7]) represent the target operational standard for future multimodal operating theaters.

Clinical Disturbances: Evaluated Gaussian noise and dropout masks model operational disturbances (lens fogging, electrocautery smoke, telemetry drops) [5]. In-vivo evaluation will further assess spontaneous bleeding and sudden camera repositioning.

VII. Conclusion

We presented a rigorous multimodal benchmark for surgical gesture recognition across all 103 JIGSAWS trials under official Leave-One-User-Out cross-validation. We confirmed the predictive primacy of normalized kinematics (H2) and the contextual value of frozen visual representations (H1). While static fusion degrades accuracy, adaptive reliability gating dynamically blends representations and enforces exact single-modality fallback under sensor loss (H4), achieving top accuracy across all three tasks. Recurrent sequence modeling suppresses oversegmentation chatter, doubling segmental Edit Scores (H5). These results demonstrate that reliability-aware dynamic fusion provides a lightweight, dependable architecture for context-aware robotic surgery.

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